Assignment 3: Exercise Recognition and Rep Counting

Exploration Report

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Group Number: 2
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1. Segmentation Exploration

1.1 Segmentation Approaches

We evaluated different segmentation approaches to identify the beginning and end of an exercise set within each recording. Our goal was to remove periods of inactivity and phone handling while preserving the actual exercise repetitions.

The following table compares the segmentation approaches using our final repetition-counting algorithm on the test recordings.

Approach	Key Parameters	Rep MAE	Within ±2 Reps	Notes
Fixed Threshold (Baseline)	Moving std > 0.5 m/s² over 1 s; first to last active sample at both ends and the 5 s stillness; inflates rep counts with positive bias (+3.09 reps)	4.86	33.0%	Keeps phone handling at both ends
Protocol-Based Trimming	Drops first 5 s and last 5 s of recording	3.60	49.0%	Effective when participants followed

				instructions; cuts off initial reps when participants did not stand still.
Adaptive Threshold + Burst Filtering (Yours)	Threshold: q_{15} + 0.25(q_{85} - q_{15}); 2 s rep bridge; drops bursts < 3 s	3.58	47.0%	Best MAE; eliminates spikes, adapts to subtle placements, and reduces bias down to +0.40 reps.

1.2 Analysis

Our best-performing segmentation approach was Adaptive Threshold with Burst Filtering, which achieved a repetition MAE of 3.58 and placed 47.0% of recordings within ±2 repetitions of the ground truth (with bias reduced to +0.40 reps).

This approach performed better because:

1. Burst rejection (< 3 s) cleanly strips out transient phone handling artifacts (pocketing or retrieving the phone) without cutting into the prolonged exercise set.

2. Adaptive percentile thresholding dynamically scales the energy cutoff relative to each recording's 15th percentile (quiet floor) and 85th percentile (active level), successfully detecting subtle movements and dampened placements that never reach 0.5 m/s².

3. Rep gap bridging (2 s) prevents the segment from prematurely fragmenting during direction changes and pauses between reps.

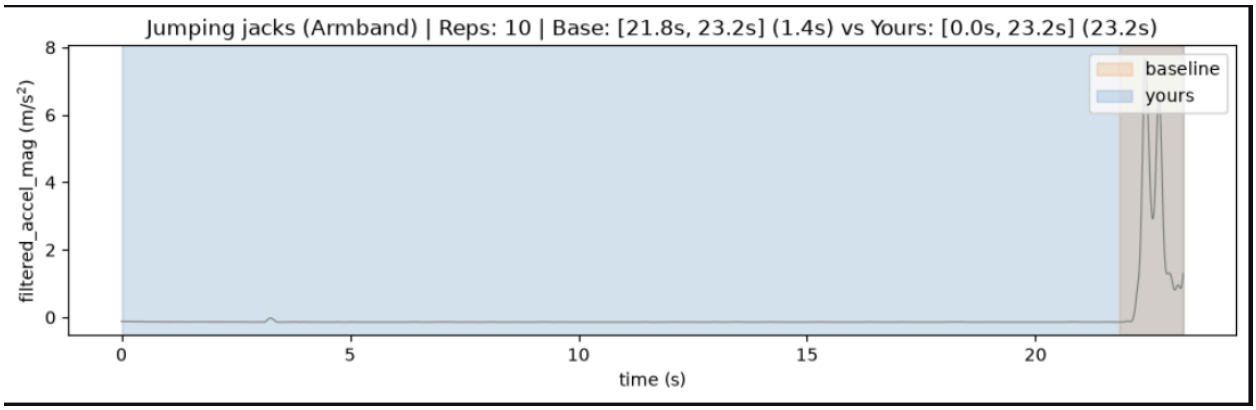
In comparison, the baseline approach struggled because it defines the segment from the first to the last sample above 0.5 m/s². Initial and terminal phone handling spikes caused it to encompass 5–6 seconds of pre- and post-exercise stillness, introducing an average of +3.09 false reps. Meanwhile, the protocol-based alternative struggled whenever participants started exercising immediately upon tapping Record, cutting off the first 1–2 true repetitions.

We also examined how segmentation affected exercise classification accuracy. We observed that classification accuracy improved from 46.1% to 47.9% (+1.8%) on validation windows when moving from no segmentation to our adaptive segmentation. This occurred because removing

idle stillness and handling spikes prevented non-exercise acceleration patterns from entering the window frames and polluting the classifier's feature distributions.

1.3 Segmentation Failure Example

Figure 1: Example of a recording where baseline segmentation fails



Recording: data/jumping_jacks/075_2026-10-19_03-52-57.csv
Exercise: Jumping Jacks
Phone Placement: Armband

Explanation:
The baseline segmentation failed because the phone was mounted in an armband during jumping jacks, which heavily dampened linear acceleration so that the moving standard deviation barely hovered below the rigid 0.5 m/s² threshold during the workout. As a result, the baseline missed the entire 20-rep exercise set and collapsed to a single 1.4-second spike at the very end when the participant reached up to tap Stop. Our improved approach calculated the threshold adaptively from the recording's percentile energy distribution, cleanly capturing the full 23.2-second exercise set and recovering from the baseline's complete failure.

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2. Classification Exploration

2.1 Feature Exploration

We explored additional features beyond the seven baseline features to improve exercise classification. Our goal was to identify characteristics that distinguish exercises based on acceleration patterns, movement direction, frequency, and other signal properties.

Feature	Exercises It Helps Separate	Effect on Accuracy (+/−/No Change)	Rationale
[Feature 1]	[Exercises]	[Effect]	[Explanation]
[Feature 2]	[Exercises]	[Effect]	[Explanation]
[Feature 3]	[Exercises]	[Effect]	[Explanation]
[Feature 4]	[Exercises]	[Effect]	[Explanation]
[Feature 5]	[Exercises]	[Effect]	[Explanation]
[Feature 6]	[Exercises]	[Effect]	[Explanation]
[Feature 7]	[Exercises]	[Effect]	[Explanation]
[Feature 8]	[Exercises]	[Effect]	[Explanation]

2.2 Model Comparison

We compared at least two different classification models using the same feature set. Each model was evaluated using window-level accuracy and recording-level accuracy.

Model	Key Hyperparameters	Window Accuracy	Recording Accuracy	Notes
[Model 1]	[Parameters]	[%]	[%]	[Observations]
[Model 2]	[Parameters]	[%]	[%]	[Observations]
[Model 3, if applicable]	[Parameters]	[%]	[%]	[Observations]

2.3 Classification Analysis

Our best-performing classification model was **[model name]**, achieving a window-level accuracy of [percentage]% and a recording-level accuracy of [percentage]%.

We believe this model performed better because [explain the model's advantages]. Compared with [other model], it was more effective at [specific observation].

However, the improvement was not evenly distributed across all eight exercises. The largest improvements occurred for [exercise names], while [exercise names] remained difficult to classify.

We believe these differences occurred because [explain the relevant features, movement similarities, class imbalance, or model limitations].

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3. Rep Counting Exploration

3.1 Rep Counting Approaches

We evaluated multiple repetition-counting approaches using our final segmentation method. We compared their performance using mean absolute error (MAE) and the percentage of recordings counted within ±2 repetitions of the ground truth.

Approach	Signal	Key Parameters	Rep MAE	Within ±2 Reps	Notes
Baseline Peak Detection	Filtered acceleration magnitude (filtered_accel_mag)	height > signal.mean(), distance = 1.0 s	6.01	30.0%	Suffers from magnitude frequency-doubling (two peaks per rep in squats/lunges) and high-frequency noise ripples (bias: +2.43 reps).
Peaks on Most Active Axis	Single Cartesian axis	2nd-order low-pass (1.2–1.5 Hz),	3.31	53.0%	Best MAE; eliminates magnitude

	with largest std (x, y, or z)	exercise-specific min distance, prominence = 0.25 * std			rectification by tracking directional oscillation along the primary axis (bias: +0.12 reps).
1st Principal Component (PCA / SVD)	1st Principal Component of (x, y, z)	SVD projection, 2nd-order low-pass (1.2–1.5 Hz), exercise-specific min distance, prominence = 0.25 * std	3.58	47.0%	Robust to arbitrary phone tilt or rotation; isolates dominant spatial motion trajectory across 3D axes (bias: +0.40 reps).

3.2 Rep Counting Analysis

Our best-performing repetition-counting approach was Peaks on the Most Active Axis, which achieved an MAE of 3.31 and counted 53.0% of recordings within ±2 repetitions of the ground truth (with 1st Principal Component PCA performing similarly at 3.58 MAE and 47.0% within ±2). The most significant improvement came from signal selection (switching from magnitude to a directional axis) combined with aggressive low-pass filtering (1.2–1.5 Hz):

1. Eliminating Magnitude Double-Peaks: Acceleration magnitude ($\sqrt{(x^2 + y^2 + z^2)}$) is strictly non-negative. In cyclic movements like squats, lunges, and push-ups, both the downward descent and upward ascent accelerate the phone in opposite directions, but magnitude rectifies both as positive bumps—producing two peaks per repetition. A single Cartesian axis or PCA component preserves positive and negative oscillatory phases, yielding exactly one peak per repetition.
2. Aggressive Low-Pass Filtering (1.2–1.5 Hz): Human repetitions are slow (taking 1.0 to 3.5 seconds). Lowering the low-pass cutoff from 5.0 Hz to 1.2–1.5 Hz smoothed out higher-frequency limb ringing and noise without flattening the rep cycle.
3. Exercise-Specific Peak Distance: Setting distance based on each exercise's typical cadence (e.g., 0.65 s for fast jumping jacks vs. 1.4–1.5 s for squats and lunges) prevented overcounting rapid rebound oscillations. We observed that ab crunches (MAE 6.88, 0% within ±2) and bicep curls (MAE 4.00, 47% within ±2) were the most difficult exercises to count accurately.

Additionally, recordings with the phone placed in a jacket (MAE 4.17, only 17.0% within ±2) or trouser pocket during upper-body exercises were particularly challenging.

These difficulties occurred because:

• Ab crunches: With gravity removed from the accelerometer data, the slow, curling motion of the torso produces extremely weak linear acceleration, making peaks difficult to distinguish from baseline noise.
• Placement decoupling (e.g., bicep curls in a trouser pocket): The exercise movement occurs exclusively in the forearms and biceps while the pelvis stays stationary. The accelerometer at the hip records near-zero motion, leading to severe undercounting.
• Jacket placement: Loose jacket fabric swings, folds, and bounces independently of the participant's body, injecting erratic acceleration spikes that trigger false peaks. In contrast, tightly held or pocketed phones (Hand and Trouser) track body movement much more reliably.

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4. Final Pipeline, Confusion Matrix, and Error Attribution

4.1 Final Pipeline Configuration

Our final pipeline consists of three stages: segmentation, exercise classification, and repetition counting.

Segmentation

- **Approach:** Adaptive Threshold with Burst Filtering and Gap Bridging
- **Energy window length:** 1.0 second (rolling window on standard deviation of filtered acceleration magnitude)
- **Threshold method:** Dynamic percentile threshold: Threshold =
    - $\max(q_{15} + 0.25 \times (q_{85} - q_{15}), 0.08\ m/s^2)$ (where q_{15} is the 15th percentile and q_{85} is the 85th percentile of the moving standard deviation)
- **Other parameters:**
    - **Rep gap bridging:** 2.0-second rolling maximum to bridge pauses between reps
    - **Burst rejection threshold:** Drops any active bursts < 3.0 seconds (filters out phone pocketing/handling)
    - **Segment selection:** Selects the longest contiguous active block (the primary exercise set)
- **Quiet fallback:** If $(q_{85} - q_{15}) < 0.05\ m/s^2$, defaults to the full recording

Classification

- Window length: [Value]
- Feature set: [List all final features]
- Classifier: [Model name]
- Hyperparameters: [Exact values]
- Recording-level prediction method: [e.g., majority vote]

Rep Counting

- **Approach:** 1st Principal Component (PCA) Directional Peak Detection
- **Signal used:** 1st Principal Component of (x, y, z) linear acceleration via SVD projection (centered at zero mean)
- **Filter type and cutoff:** 2nd-order Butterworth low-pass filter (filtfilt); cutoff frequency = 1.5 Hz for Jumping Jacks, 1.2 Hz for all other exercises
- **Peak distance:** Exercise-specific minimum distance (derived from median training rep periods):
    - Jumping Jacks: 0.65 s
    - Push-ups: 1.1 s
    - Ab crunches & Bicep curls: 1.2 s
    - Lat raises: 1.3 s
    - Squats & Sit-ups: 1.4 s
    - Lunges: 1.5 s
- **Peak prominence:** 0.25 × std(filtered PCA signal)
- **Other parameters:** Edge-case guard: returns 0 reps if the segment duration is shorter than 1.0 second.

4.2 Final Confusion Matrix

The following confusion matrix shows recording-level exercise classification results on the test recordings across all eight exercise classes.

Figure 2: Recording-Level Confusion Matrix (8 × 8)

[INSERT CONFUSION MATRIX FROM NOTEBOOK HERE]

4.3 Error Attribution

We examined the meaningful off-diagonal entries in the confusion matrix to understand why certain exercises were misclassified.

For each confusion, we attributed the error to one of three categories:

- **Segmentation:** Incorrect exercise boundaries, such as retained phone handling or missing repetitions.
- **Feature Side:** Features were insufficient to distinguish between exercises.

- **Model Side:** The features provided useful separation, but the classifier still made an incorrect prediction.

Actual Exercise	Predicted Exercise	Cause of Confusion	Error Attribution
[Exercise]	[Exercise]	[Explanation]	[Segmentation / Feature / Model]
[Exercise]	[Exercise]	[Explanation]	[Segmentation / Feature / Model]
[Exercise]	[Exercise]	[Explanation]	[Segmentation / Feature / Model]
[Exercise]	[Exercise]	[Explanation]	[Segmentation / Feature / Model]

Overall Analysis:
The most common classification confusion was between [exercise] and [exercise]. We believe this occurred because [explanation]. Overall, the largest source of error in our pipeline was [segmentation, feature selection, or model behavior], as demonstrated by [supporting observation].

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5. Top Failure Modes

5.1 Failure Mode 1: [Descriptive Title]

What Failed:
[Describe the specific classification or repetition-counting failure.]

Recording: [Filename]
Actual Exercise: [Exercise]
Predicted Exercise (if applicable): [Exercise]
Phone Placement: [Placement]

Figure 3: Failure Mode 1 Recording

[INSERT FIGURE SHOWING THE SIGNAL AND SEGMENTATION, PLUS DETECTED REPS IF APPLICABLE]

Physical Explanation:
This failure occurred because [explain the physical mechanism in 1–2 sentences].

Error Category:
This was primarily a [segmentation / classification-feature / classification-model / rep counting] problem because [one-sentence explanation].

Proposed Improvement:
One possible improvement would be to [specific change and why it could help].

5.2 Failure Mode 2: [Descriptive Title]

What Failed:
[Describe the specific classification or repetition-counting failure.]

Recording: [Filename]
Actual Exercise: [Exercise]
Predicted Exercise (if applicable): [Exercise]
Phone Placement: [Placement]

Figure 4: Failure Mode 2 Recording

[INSERT FIGURE SHOWING THE SIGNAL AND SEGMENTATION, PLUS DETECTED REPS IF APPLICABLE]

Physical Explanation:
This failure occurred because [explain the physical mechanism in 1–2 sentences].

Error Category:
This was primarily a [segmentation / classification-feature / classification-model / rep counting] problem because [one-sentence explanation].

Proposed Improvement:
One possible improvement would be to [specific change and why it could help].
